

Permanents & Expected Permutations

The **permanent** of a matrix is an analogue to its determinant: A *multilinear function of the elements which produces a scalar*.

Unlike the determinant, the permanent is a famously computationally-complex mapping, with the naïve algorithm requiring $O(N!)$ operations, and the best general-purpose algorithm requiring $O(2^N * N^2)$.

The **expected permutation** of a cost matrix A describes the expected matching of resources to tasks if this assignment is made proportional to its cost. *Ex. Anticipated One-to-One Resource Allocation*

Computation of $E\{A\}$ is **directly related** to computing the permanent and is hence computationally intensive.

$$perm(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i,\sigma(i)}$$

$$E(A) = \frac{1}{perm(A)} \sum_{\sigma \in S_n} M_{n \times n}^{\sigma} \prod_{i=1}^n a_{i,\sigma(i)} \Rightarrow E_{ij} = \frac{a_{ij} perm(A_{ij})}{perm(A)}$$

How can perm(A) and E(A) be effectively approximated in polynomial time?

Methodology

- By iteratively scaling the rows and columns of a matrix A, the **Sinkhorn Decomposition** produces a **Doubly Stochastic**, the same matrix family as the expected permutation.

$$Sinkhorn\{A\} = D_1 \Omega D_2$$

For doubly stochastic Ω and diagonal matrices D .

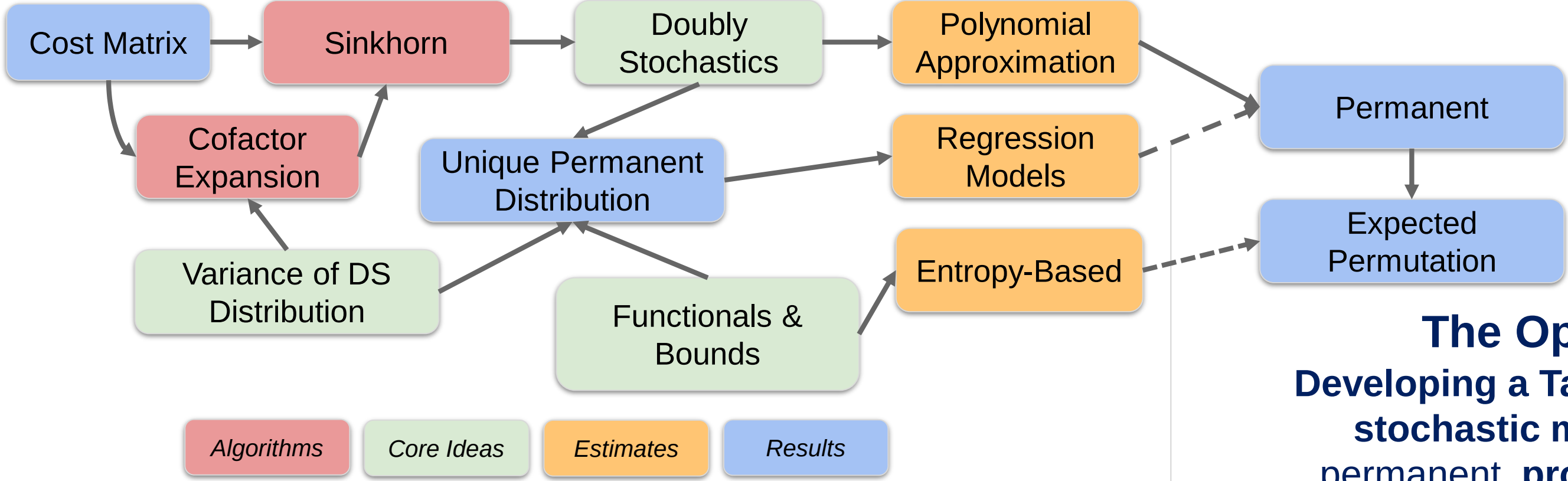
- This produces a class of efficient approximation schemes;

$$E(A) = Sinkhorn\{a_{ij} \prod_{l=1}^n D_{1l} D_{2l} perm(\Omega_{ij}) ; i, j \in [1, n]\}$$

$$\approx Sinkhorn\{a_{ij} \prod_{l=1}^n D_{1l} D_{2l} \theta(\Omega_{ij})\}$$

- To find θ , it is now pertinent to **approximate the permanent of a Doubly Stochastic matrix**.

Algorithmic Flow-Chart



99.88%-99.95% Accuracy in $\sim O(N^9)$

The Optimal Approach: Taylor Series
 Developing a Taylor Series for the permanent of a doubly stochastic matrix, centered at the matrix of minimum permanent, **produced both elegant and highly effective approximations**.

$$perm(X_{n \times n}) = \frac{n!}{n^n} + \frac{1}{2!} \frac{(n-2)!}{n^{n-2}} \sum_{i,j} (x_{ij} - \frac{1}{n})^2 + \frac{4}{3!} \frac{(n-3)!}{n^{n-3}} \sum_{i,j} (x_{ij} - \frac{1}{n})^3$$

$$+ \frac{1}{4!} \frac{(n-4)!}{n^{n-4}} (36 \sum_{i,j} (x_{ij} - \frac{1}{n})^4 + \psi_4(X)) + O(x_{ij} - 1/n)^5 \dots$$

Higher order terms become **increasingly more intricate**, however can be further **estimated**, via an inclusion-exclusion argument, as a series of ‘**moments**’.

$$T_n(X) = \frac{n!}{n^n} + \frac{n!}{n^n} \sum_{k=2}^{n-1} \frac{(k-1)^2}{k!^2 \binom{n}{k}} n^k \sum_{i,j} (x_{ij} - \frac{1}{n})^k$$

Avenues of Approximation

Numerous possibilities for theta were analyzed over the course of the project.

$$\theta : \Omega_{n \times n} \rightarrow \approx perm \in \mathbb{R}$$

Initial experimentation found success with:

- $perm \approx \frac{n!}{n^n}$ (Lower Bound)
- $\ln(perm)$ regression with density and entropy parameters
- $\theta = \text{Constant across matrix minors}$
- $\theta = Var(A)$

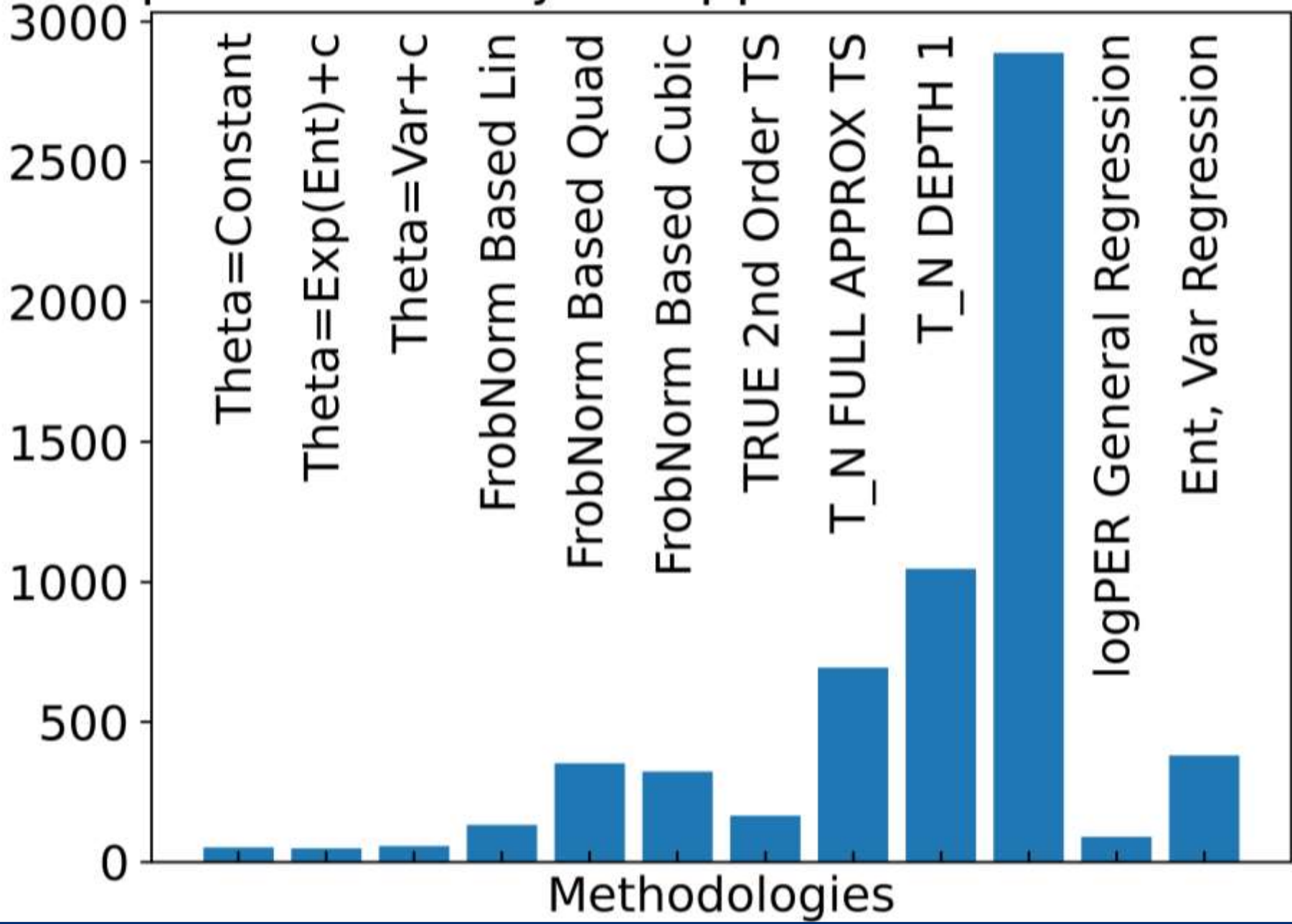
Work by **Bregman (1973)** indicated:

- $perm \propto \exp(Entropy)$
- KL Divergence to Van der Waerden Matrix

Future Goals

- Rigorous **bounds** on the maximum approximation error or on the expected deviance from the true permanent.
- Tailored approximations** for matrices of a particular **density, size, and makeup**.
- Generalize** the elements of the **Taylor Series** using partitions to compute past the quartic term.

Comparison of Project Approximation Techniques.



Supplemental Work

- Developed **novel algorithms** for the **explicit computation** of the permanent & **matrix-multiplication representations** for several permanent algorithms.
- Produced **statistical moments** for the permanent of uniform matrices.
- Formed a **probability distribution** of elements of matrices under the Sinkhorn transformation.
- Investigated the **uniform sampling of Doubly Stochastic** matrices from points on a hypersphere.
- Formed **inequalities** for the permanent of positive matrices.